

FedGATSage vs. centralised XGBoost/RF on NF-ToN-IoT and NF-ToN-IoT-I

FedHetFormer-IDS experiment log §11p and §11r · 27 September 2026

Setup. The same 1500-rows-per-label sets were used for every run (seed 42; mitm 1,295, ransomware 142 in NF; scanning 1,460, dos 283, ransomware 142 in FHF). All 10 labels, no payload. FedGATSage: `asfi50/Fed_GNN` branch v2, 5 clients, 15 rounds; test $\approx 10\%$ (1,344 / 1,239 flows). Centralised: branch `centralised` scripts, per-flow features only (IPs and source port dropped), `--samples-per-class 1500 --min-row-count 1`; test 20% split by flow vector (2,851 / 3,248 flows). FHF = NF-ToN-IoT v1 with `Attack/Label` taken from matched ToN-IoT PCAP/Zeek flows (Kaggle `mtasfi/nftoniot-fhf`).

1. Overall (all 10 labels)

	FedGATSage NF-ToN-IoT	FedGATSage FHF	XGBoost FHF	RF FHF	XGBoost NF-ToN-IoT	RF NF-ToN-IoT
Accuracy	0.434	0.616	0.752	0.748	0.701	0.702
Balanced acc.	0.484	0.516	0.713	0.710	0.695	0.696
Macro-F1	0.448	0.474	0.709	0.710	0.701	0.700

2. F1 (L7 classes)

	FedGATSage NF-ToN-IoT	FedGATSage FHF	XGBoost FHF	RF FHF	XGBoost NF-ToN-IoT	RF NF-ToN-IoT
injection	0.36	0.34	0.47	0.41	0.46	0.50
password	0.31	0.75	0.56	0.53	0.35	0.37
xss	0.00	0.02	0.50	0.49	0.41	0.38

3. Recall (L7 classes)

	FedGATSage NF-ToN-IoT	FedGATSage FHF	XGBoost FHF	RF FHF	XGBoost NF-ToN-IoT	RF NF-ToN-IoT
injection	0.47	0.38	0.40	0.36	0.43	0.47
password	0.20	0.76	0.59	0.51	0.35	0.37
xss	0.00	0.01	0.49	0.51	0.46	0.39

4. Precision (L7 classes)

	FedGATSage NF-ToN-IoT	FedGATSage FHF	XGBoost FHF	RF FHF	XGBoost NF-ToN-IoT	RF NF-ToN-IoT
injection	0.29	0.31	0.57	0.48	0.49	0.53
password	0.67	0.74	0.54	0.54	0.35	0.37
xss	0.00	0.10	0.51	0.47	0.37	0.36

Reading.

- With equal settings, the centralised flow-only models clearly beat FedGATSage: macro-F1 ≈ 0.71 against 0.45–0.47.
- L7 (injection, password, xss) is the weakest group in every run. The only L7 F1 above 0.56 is FedGATSage on FHF password (0.75), and the same model gets xss almost entirely wrong on both datasets.
- FHF helps the centralised models on password (+0.18–0.21 F1) and xss (+0.09–0.11), but overall macro-F1 by only ≈ 0.01 .
- One seed, and 140–330 test flows per L7 class. The FedGATSage and centralised test splits differ, so differences of 1–2 points are noise. The gap of 0.71 against 0.47 is not.